

Phase 04 — Execution · eCRF Pictorial walkthrough (X1: Form Builder + X2: EDC capture)

Phase context. Once the trial is approved and registered, real subjects enrol and real data accumulates. The eCRF subsystem is the platform's regulatory-grade data-capture core (Part 11 / ALCOA+ / GCP / ICH E6). This guide walks the two foundational demos: **X1 — design CRFs from a protocol** (admin / data-manager), and **X2 — capture subject data with edit-checks + queries** (site coordinator).

Captured against a **live instance** (`http://localhost:8000` , Bedrock `claude-haiku-4-5`) on 2026-06-19. The two eCRF pages are plain vanilla JS (no React/Babel), so they are unaffected by the chat-UI Babel issue noted in the Phase 02 guide.

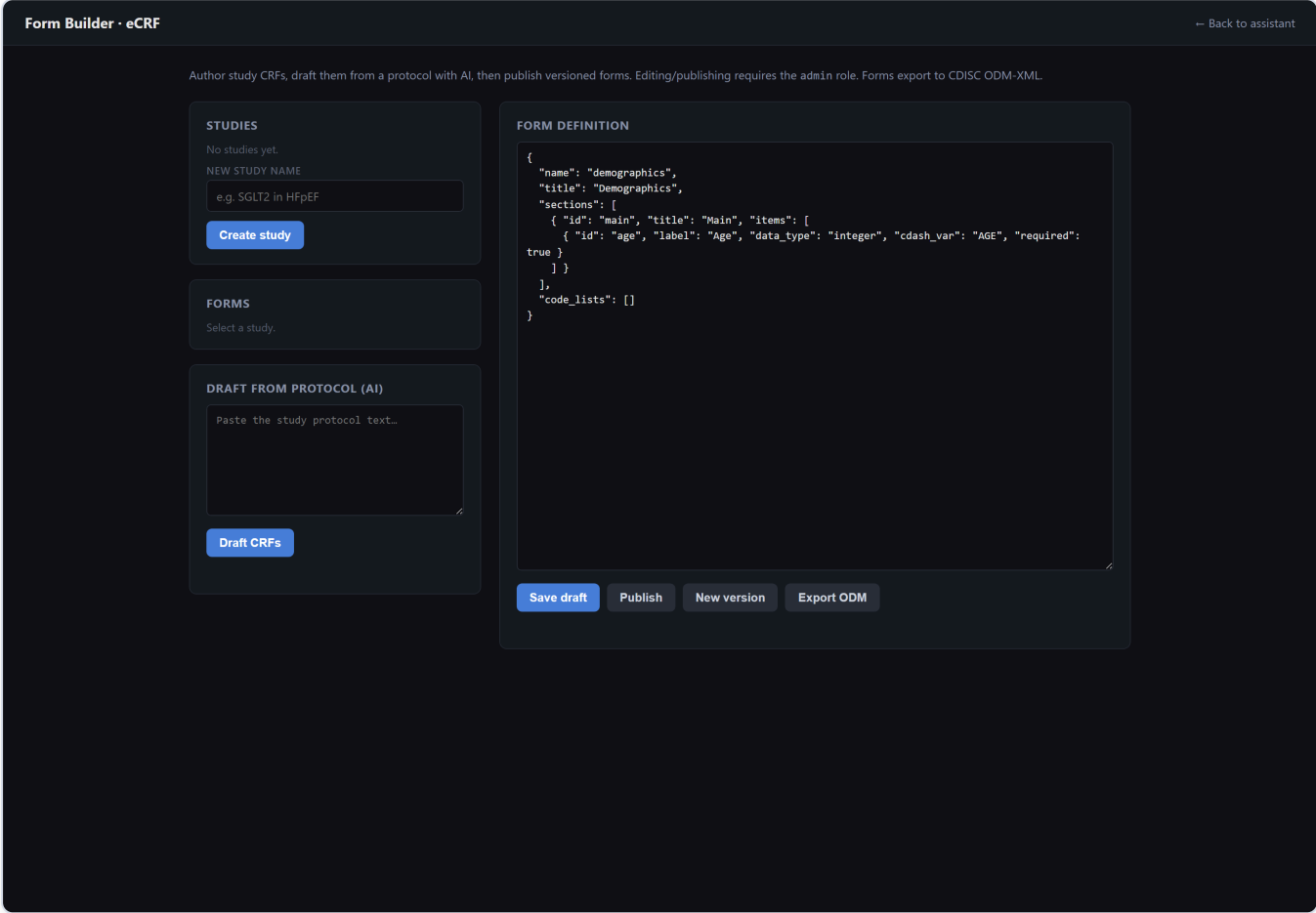
Surfaces used:

`http://localhost:8000/ecrf.html` — Form Builder (X1)
`http://localhost:8000/collector.html` — Site EDC capture (X2)

X1 — eCRF design (AI-draft CRFs from a protocol)

Step 1 — Open the Form Builder

`http://localhost:8000/ecrf.html` . Left column = Studies / Forms / AI draft; right column = the form-definition editor.



Step 2 — Create the study

Type a name and click **Create study**, then select it. (We used `SMOKE-T2DM-001 — Phase 2 SGLT2i in adults with T2DM (smoke study)`.)

STUDIES

SMOKE-T2DM-001 — Phase 2 SGLT2i in
adults with T2DM (smoke study)

NEW STUDY NAME

e.g. SGLT2 in HFpEF

Create study

Step 3 — Paste the protocol into "Draft from protocol (AI)"

DRAFT FROM PROTOCOL (AI)

Patient-reported outcomes (ePRO, weekly):

- Diabetes Treatment Satisfaction Questionnaire (DTSQ, 8 items).
- Self-reported hypoglycaemia events (frequency + severity).

Draft CRFs

▼ Protocol text used

Phase 2, single-arm, open-label study of an SGLT2 inhibitor in adults with type 2 diabetes mellitus.

Eligibility:

- Adults aged 18-75 years.
- HbA1c 7.0-10.0% at screening.
- On stable metformin ≥ 3 months prior to enrolment.
- BMI 22-40 kg/m².

Schedule of assessments:

- Visit 1 (Baseline, Week 0): consent, demographics, vital signs, fasting glucose, HbA1c, body weight, eligibility.
- Visit 2 (Week 4): vital signs, fasting glucose, AE review.
- Visit 3 (Week 12): vital signs, fasting glucose, HbA1c, body weight, AE review, lab safety panel.
- Visit 4 (Week 24, primary endpoint): vital signs, fasting glucose, HbA1c, body weight, AE review, lab safety panel.
- Visit 5 (EOS, Week 28): final vital signs, AE review, treatment-emergent serious AE inventory.

Primary outcome: change in HbA1c from baseline to Week 24.

Secondary outcomes: fasting glucose, body weight, proportion with HbA1c $< 7.0\%$ at Week 24.

Safety outcomes: vital signs, lab safety panel, treatment-emergent adverse events (graded per CTCAE v5).

Patient-reported outcomes (ePRO, weekly):

- Diabetes Treatment Satisfaction Questionnaire (DTSQ, 8 items).
- Self-reported hypoglycaemia events (frequency + severity).

Step 4 — Draft the CRFs

Click **Draft CRFs** (~10–40 s on Bedrock). The model proposes a full form set — Demographics, Eligibility, Vital Signs, Lab Safety, Body Weight/BMI, AE Reporting, Study-Drug, DTSQ ePRO, Hypoglycaemia. **Nothing auto-saves.**

DRAFT FROM PROTOCOL (AI)

Patient-reported outcomes (ePRO, weekly):

- Diabetes Treatment Satisfaction Questionnaire (DTSQ, 8 items).
- Self-reported hypoglycaemia events (frequency + severity).

Draft CRFs

Drafted 9 form(s). **Assumptions & Reviewer Checklist:** 1. **DTSQ & Hypoglycaemia (ePRO):** Mapped as weekly assessments in the protocol but no explicit event structure provided. Assumed these are collected via patient portal at Week 0, 4, 12, 24, and 28 (aligned with visit schedule). Review timing and ensure ePRO platform supports weekly cadence. 2. **Safety Lab Panel:** "Lab safety panel" not specified in detail. Included standard hepatic and renal markers (ALT, AST, creatinine, eGFR) typical for SGLT2i safety monitoring. Protocol reviewer should confirm exact lab analytes required. 3. **Fasting Glucose Units:** Assumed mg/dL; confirm with protocol or local convention if mmol/L preferred. 4. **CTCAE v5 Grading:** Applied to all AEs per protocol requirement. Some SGLT2i-related AEs (e.g., genital infections, DKA) may warrant additional

targeted items—consider adding conditional branching or a separate form if needed. 5. ****Study Drug Dispensing:**** Created at Baseline only; protocol does not mention re-dispensing at later visits. If drug is re-dispensed, extend collection to relevant visits. 6. ****Edit Checks:**** Added conservative soft/hard checks for vital signs, labs, and demographics ranges. Severity="soft" used for plausible physiologic ranges to minimize false blocks; adjust thresholds as needed. 7. ****Missing Data Points:**** Protocol mentions "HbA1c <7.0% at Week 24" as a secondary outcome (proportion) but this is derived, not collected. The primary form collects HbA1c value; analysis will compute proportion. 8. ****Eligibility Verification at Baseline:**** Created a dedicated form to document screening/eligibility criteria and enrollment confirmation. Some sites may prefer this as a pre-baseline "Screening" visit. 9. ****DTSQ Items:**** Listed as "8 items" in protocol; DTSQ standard version varies. Included 8 representative satisfaction/convenience/recommendation items. Confirm exact items and scoring scale with validated instrument. 10. ****No Concomitant Medication Form:**** Protocol does not explicitly require detailed concomitant medication CRF (only notes "on stable metformin ≥ 3 months"). Consider adding if medication changes during study are tracked. 11. ****Visit Windows:**** Applied standard visit windows (± 7 days for Week 4, ± 4 weeks for Week 12/24, flexible EOS). Adjust per protocol CRF guidance.

Demographics (demographics)

load →

Enrollment & Eligibility Verification
(enrollment)

load
→

Vital Signs (vital_signs)

load →

Anthropometry (anthropometry)

load →

Laboratory Assessments (laboratory)

load →

Adverse Events (adverse_events)

load →

Study Drug Dispensing (study_drug)

load →

Diabetes Treatment Satisfaction
Questionnaire (DTSQ) (dtsq)

load
→

Self-Reported Hypoglycaemia
(hypoglycemia)

load
→

Step 5 — Load a draft into the editor

Click a drafted form (here **Vital Signs**) to load its JSON. Note the CDASH naming (`cdash_var: VSORRES / SYSBP`), the structured items, and the AI's starting edit-check.

FORM DEFINITION · (NEW DRAFT FROM AI — REVIEW THEN SAVE)

```
{
  "name": "vital_signs",
  "title": "Vital Signs",
  "sections": [
    {
      "id": "vitals_section",
      "title": "Vital Signs",
      "items": [
        {
          "id": "sys_bp",
          "label": "Systolic Blood Pressure",
          "data_type": "integer",
          "cdash_var": "SYSBP",
          "required": true,
          "units": "mmHg",
          "code_list_ref": null,
          "edit_checks": [
            {
              "id": "sys_bp_range",
              "severity": "soft",
              "expression": "is_blank(sys_bp) or (sys_bp >= 70 and sys_bp <= 250)",
              "message": "Systolic BP should be between 70 and 250 mmHg"
            }
          ],
          "help_text": null
        }
      ]
    }
  ]
}
```

Save draft

Publish

New version

Export ODM

Loaded AI draft into the editor. Review, then Save to create it under the selected study.

Step 6 — Designer review: harden an edit-check (human-in-the-loop)

The designer reviews and **edits** before publishing. Here we upgrade the systolic-BP range check to **hard** (blocks save) and add a **soft** >180 warning (auto-raises a query). This demonstrates that the AI draft is a starting point, not an auto-published artefact.

FORM DEFINITION · (NEW DRAFT FROM AI — REVIEW THEN SAVE)

```
{
  "name": "vital_signs",
  "title": "Vital Signs",
  "sections": [
    {
      "id": "vitals_section",
      "title": "Vital Signs",
      "items": [
        {
          "id": "sys_bp",
          "label": "Systolic Blood Pressure",
          "data_type": "integer",
          "cdash_var": "SYSBP",
          "required": true,
          "units": "mmHg",
          "code_list_ref": null,
          "edit_checks": [
            {
              "id": "sys_bp_range_hard",
              "severity": "hard",
              "expression": "is_blank(sys_bp) or (sys_bp >= 70 and sys_bp <= 250)",
              "message": "Systolic BP must be between 70 and 250 mmHg (hard limit)."
            },
            {
              "id": "sys_bp_high_soft",
              "severity": "soft",
              "expression": "is_blank(sys_bp) or (sys_bp <= 180)"
            }
          ]
        }
      ]
    }
  ]
}
```

Save draft

Publish

New version

Export ODM

Loaded AI draft into the editor. Review, then Save to create it under the selected study.

Step 7 — Save the draft, then Publish (immutable)

Save draft → form lands at `status=draft`. **Publish** → `draft` → `published`; the definition is now immutable (further edits require **New version**).

FORM DEFINITION · (NEW DRAFT FROM AI — REVIEW THEN SAVE)

```
{
  "name": "vital_signs",
  "title": "Vital Signs",
  "sections": [
    {
      "id": "vitals_section",
      "title": "Vital Signs",
      "items": [
        {
          "id": "sys_bp",
          "label": "Systolic Blood Pressure",
          "data_type": "integer",
          "cdash_var": "SYSBP",
          "required": true,
          "units": "mmHg",
          "code_list_ref": null,
          "edit_checks": [
            {
              "id": "sys_bp_range_hard",
              "severity": "hard",
              "expression": "is_blank(sys_bp) or (sys_bp >= 70 and sys_bp <= 250)",
              "message": "Systolic BP must be between 70 and 250 mmHg (hard limit).",
            },
            {
              "id": "sys_bp_high_soft",
              "severity": "soft",
              "expression": "is_blank(sys_bp) or (sys_bp <= 180)"
            }
          ]
        }
      ]
    }
  ]
}
```

Save draft

Publish

New version

Export ODM

Published v1.

FORMS · SMOKE-T2DM-001 — PHASE 2 SGLT2I IN ADULTS WITH T2DM (SMOKE STUDY)

vital_signs

v1

published

Step 8 — Export CDISC ODM-XML

Export ODM downloads a CDISC ODM v1.3.2 document ([phase04-ecrf/form.odm.xml](#)) with `Study` + `FormDef` + `ItemDef` s.

X1 sanity checks

Check	Expected	Observed
Nothing auto-publishes	Drafted forms stay until you Save	✓
CDASH naming	Demographics/VS fields follow CDASH (VSORRES , SYSBP ...)	✓
Hard vs soft checks distinct	Hard blocks; soft warns	✓ (set in Step 6)
Published forms immutable	PUT on a published form → 409; must New-version	✓
ODM export	Valid CDISC ODM-XML	✓ v1.3.2

X2 — EDC capture (site coordinator)

Pre-req: X1 published the Vital Signs form. A deployment snapshots published form definitions.

Note: `collector.html` has **no "New deployment" button** — deployments are created via the EDC API (`POST /api/edc/deployments` with the published `research_study_id`). The capture script creates the *Smoke deployment* this way, then drives the page. Everything below is the coordinator's day-to-day surface.

Step 1 — Open the collector, select the deployment

`http://localhost:8000/collector.html` → pick **Smoke deployment** in the Deployment dropdown.

Data Capture · EDC

← Back to assistant

Enter subject data against deployed CRFs. Hard edit-checks block save; soft checks raise queries. All entries are audited.

DEPLOYMENT

— select —

SITES & SUBJECTS

New site name

+ Site

ADD SUBJECT

Subject code e.g. S-001

+ Subject

OPEN A FORM

Open form for subject

Select a subject.

SUBMISSIONS (CDISC)

Select a deployment to view CDISC datasets.

RECRUITMENT

Select a deployment to view the recruitment funnel.

VISIT CALENDAR

Select a deployment to view the visit calendar.

SOURCE EXTRACTION

FORM

Open or select a form instance.

QUERIES

—

SAFETY (AES + DEVIATIONS)

Select a subject to load safety records.

Step 2 — Add a site and a subject

Add site `Site 01 — University Hospital`, then add subject `S01-001` against that site.

SITES & SUBJECTS

New site name

+ Site

ADD SUBJECT

Site 01 — University Hospital

Subject code e.g. S-001

+ Subject

S01-001

Step 3 — Open the Vital Signs form for the subject

Select the subject, pick **Vital Signs** in the deployed-forms dropdown, and **Open form for subject** → creates a `FormInstance` at `status=in_progress`.

FORM · VITAL SIGNS V1 (BLANK)

Vital Signs

Systolic Blood Pressure *

Diastolic Blood Pressure *

Heart Rate *

Body Temperature

☐

mark complete

Reason for change (on edits)

Save

Step 4 — Enter valid data → save succeeds

Systolic 120 / Diastolic 78 / Heart rate 72 → **Save** → persists to `ItemData` (status saved).

FORM · VITAL SIGNS V1 (BLANK)

Vital Signs

Systolic Blood Pressure *

120

Diastolic Blood Pressure *

78

Heart Rate *

72

Body Temperature

☐

mark complete

Reason for change (on edits)

Save

Saved (status: in_progress).

Step 5 — Hard edit-check → save blocked (422)

Set systolic BP to **300** (outside the hard 70–250 limit) → **Save** → **422, nothing persisted**. The offending field shows a red inline failure and the status reads *"Hard check(s) failed — nothing saved."*

FORM · VITAL SIGNS V1 (BLANK)

Vital Signs

Systolic Blood Pressure *

300

Systolic BP must be between 70 and 250 mmHg (hard limit).

Diastolic Blood Pressure *

78

Heart Rate *

72

Body Temperature

☐ mark complete

Re-entry of systolic BP reading.

Save

Hard check(s) failed — nothing saved. Fix the flagged fields.

Step 6 — Soft edit-check → saves and auto-raises a query

Correct to systolic BP **200** (within the hard limit, but above the soft 180 threshold) → **Save** → data **saves** and an **auto query** appears in the Queries panel referencing the offending `item_id`.

QUERIES

sys_bp — Systolic BP above 180 mmHg — verify reading (possible hypertensive value).

auto · open

response

Respond

Close

From here the coordinator/monitor can **Respond** or **Close** the query — the full query-lifecycle.

X2 sanity checks

Check	Expected	Observed
Hard check	Blocked save, 422, no <code>ItemData</code> written	✅ BP 300 → 422
Soft check	Saves and auto-raises a query referencing the <code>item_id</code> + rule	✅ BP 200 → <code>sys_bp</code> query
Audit trail	Every save records <code>AuditEntry</code> (old→new, actor, source=edc)	inspect <code>GET /api/edc/form-instances/{id}/audit</code>

Reproduce

```
uv run python scripts/demo_capture_x1_ecrf.py    # X1 -> phase04-ecrf/00..08 + form.odm.xml
uv run python scripts/demo_capture_x2_edc.py     # X2 -> phase04-ecrf/10..16
```

The AI assigns slightly different item ids each run (`sbp` / `sys_bp` / `systolic_bp`). The X1 script hardens the systolic check by **label match** ("Systolic"), and X2 discovers field ids from the deployed form at runtime — so both scripts are robust to that non-determinism.